G:/code/vhdl/simple_traffic.vhd (/tb_simple_traffic/uut) - Default

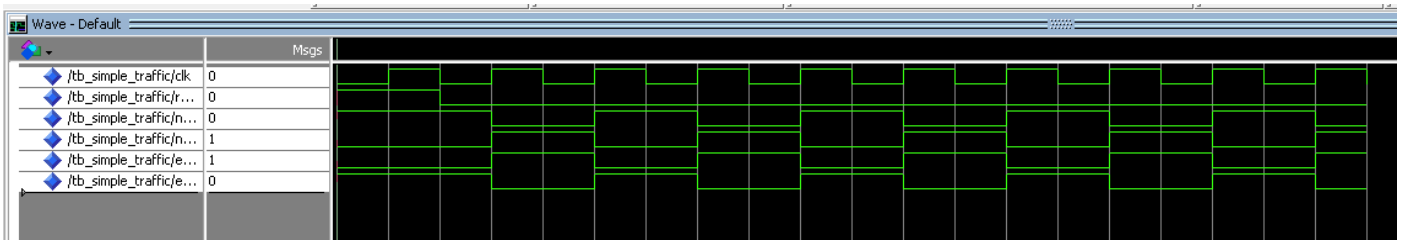
```
Ln#
 1  library ieee;
 2  use ieee.std_logic_1164.all;
 3
 4  entity simple_traffic is
 5      port (
 6          clk      : in  std_logic;
 7          reset    : in  std_logic;
 8          ns_green : out std_logic;
 9          ns_red   : out std_logic;
10          ew_green : out std_logic;
11          ew_red   : out std_logic
12      );
13  end simple_traffic;
14
15  architecture behavioral of simple_traffic is
16
17      type state_type is (ns_go, ew_go);
18      signal state : state_type := ns_go;
19
20  begin
21
22      process(clk, reset)
23      begin
24          if reset = '1' then
25              state <= ns_go;
26          elsif rising_edge(clk) then
27              case state is
28                  when ns_go =>
29                      state <= ew_go;
30
31                  when ew_go =>
32                      state <= ns_go;
33              end case;
34          end if;
35      end process;
36
37      process(state)
38      begin
39          case state is
40              when ns_go =>
41                  ns_green <= '1';
42                  ns_red   <= '0';
43                  ew_green <= '0';
44                  ew_red   <= '1';
45
46              when ew_go =>
47                  ns_green <= '0';
48                  ns_red   <= '1';
49                  ew_green <= '1';
50                  ew_red   <= '0';
51          end case;
52      end process;
53
54  end behavioral;
```

Program 1: VHDL Code for Simple Traffic Light Controller

G:/code/vhdl/tb_simple_traffic.vhd (/tb_simple_traffic) - Default

```
Ln#
 1  library ieee;
 2  use ieee.std_logic_1164.all;
 3
 4  entity tb_simple_traffic is
 5  end tb_simple_traffic;
 6
 7  architecture test of tb_simple_traffic is
 8
 9      signal clk : std_logic := '0';
10      signal reset : std_logic := '0';
11      signal ns_green, ns_red, ew_green, ew_red : std_logic;
12
13  begin
14
15      uut: entity work.simple_traffic
16          port map (
17              clk => clk,
18              reset => reset,
19              ns_green => ns_green,
20              ns_red => ns_red,
21              ew_green => ew_green,
22              ew_red => ew_red
23          );
24
25      clk <= not clk after 5 ns;
26
27      process
28      begin
29          reset <= '1';
30          wait for 10 ns;
31
32          reset <= '0';
33          wait for 40 ns;
34
35          wait;
36      end process;
37
38  end test;
39
```

Program 2: VHDL Testbench for Simple Traffic Light Controller



Testbench Output: Simulation Waveform of Simple Traffic Light Controller